BIOS 731: HW3

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Problem 0

GitHub Repo: https://github.com/rndparr/bios731_hw3_parrish

Problem 1: Newton's method

- Derive likelihood, gradient, and Hessian for logistic regression for an arbitrary number of predictors p

Let $\boldsymbol{\theta}$ be a $p \times 1$ vector of parameters.

$$\pi_i = \text{logit}^{-1}(\mathbf{X}_i^T\boldsymbol{\theta}) = \frac{\exp[\mathbf{X}_i^T\boldsymbol{\theta}]}{1+\exp[\mathbf{X}_i^T\boldsymbol{\theta}]}$$

$$= \frac{1}{1+\exp[-(\mathbf{X}_i^T\boldsymbol{\theta})]}$$

$$\begin{aligned}
\mathcal{L}(\pi_i) &= \prod_{i=1}^{n} \pi_i^{Y_i}(1-\pi_i)^{1-Y_i} \\
l(\pi_i) &= \sum_{i=1}^{n} Y_i \log(\pi_i) + (1-Y_i)\log(1-\pi_i) \\
&= \sum_{i=1}^{n} Y_i \left[\log(\pi_i) - \log(1-\pi_i)\right] + \log(1-\pi_i) \\
&= \sum_{i=1}^{n} Y_i \log\left(\frac{\pi_i}{1-\pi_i}\right) + \log(1-\pi_i) \\
&= \sum_{i=1}^{n} Y_i \,\text{logit}(\pi_i) + \log(1-\pi_i) \\
l(\boldsymbol{\theta}) &= \sum_{i=1}^{n} Y_i\mathbf{X}_i^T\boldsymbol{\theta} + \log\left(1 - \frac{\exp[\mathbf{X}_i^T\boldsymbol{\theta}]}{1+\exp[\mathbf{X}_i^T\boldsymbol{\theta}]}\right) \\
&= \sum_{i=1}^{n} Y_i\mathbf{X}_i^T\boldsymbol{\theta} + \log\left(\frac{1}{1+\exp[\mathbf{X}_i^T\boldsymbol{\theta}]}\right) \\
&= \sum_{i=1}^{n} Y_i\mathbf{X}_i^T\boldsymbol{\theta} - \log\big(1+\exp[\mathbf{X}_i^T\boldsymbol{\theta}]\big)
\end{aligned}$$

The gradient is:

$$\nabla l(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left(Y_i - \frac{\exp[\mathbf{X}_i^T \boldsymbol{\theta}]}{1 + \exp[\mathbf{X}_i^T \boldsymbol{\theta}]} \right) \mathbf{X}_i$$

$$\nabla l(\pi_i) = \sum_{i=1}^{n} (Y_i - \pi_i) \mathbf{X}_i$$

The Hessian is:

$$\nabla^2 l(\boldsymbol{\theta}) = -\sum_{i=1}^{n} \frac{\exp[\mathbf{X}_i^T \boldsymbol{\theta}]}{\left\{1 + \exp[\mathbf{X}_i^T \boldsymbol{\theta}]\right\}^2} \mathbf{X}_i \mathbf{X}_i^T$$

$$\nabla^2 l(\pi_i) = -\sum_{i=1}^{n} \pi_i (1 - \pi_i) \mathbf{X}_i \mathbf{X}_i^T$$

- What is the Newton's method update for β for logistic regression?

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \left[\nabla^2 l(\boldsymbol{\theta}^{(k)}) \right]^{-1} \nabla l(\boldsymbol{\theta}^{(k)})$$

- Is logistic regression a convex optimization problem? Why or why not?

Yes; the goal is to find a global optimum that minimizes the differentiable convex function $l(\boldsymbol{\theta})$.

Problem 2: MM

(A) In constructing a minorizing function, first prove the inequality

$$-\log\{1 + \exp x_i^T \theta\} \geq -\log\{1 + \exp(X_i^T \theta^{(k)})\} - \frac{\exp(X_i^T \theta) - \exp(X_i^T \theta^{(k)})}{1 + \exp(X_i^T \theta^{(k)})}$$

with equality when $\theta = \theta^{(k)}$. This eliminates the log terms.

Answer:

When $\theta = \theta^{(k)}$, the second term on the right hand side becomes 0, hence equality.

If $f(y)$ is convex and differentiable, then $f(y) \geq f(x) + f'(x)(y - x)$. The function $-\log\{x\}$ is convex and its derivative is $-\frac{1}{x}$. Then

$$\begin{aligned} -\log\left\{1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta})\right\} &\geq -\log\left\{1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})\right\} - \frac{1}{1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})} \left\{ \left[1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta})\right] - \left[1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})\right] \right\} \\ &= -\log\left\{1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})\right\} - \frac{\exp\left(\mathbf{X}_i^T \boldsymbol{\theta}\right) - \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})}{1 + \exp(\mathbf{X}_i^T \boldsymbol{\theta}^{(k)})} \end{aligned}$$

(B) Now apply the arithmetic-geometric mean inequality to the exponential function $\exp(X_i^T \theta)$ to separate the parameters. Assuming that θ has p components and that there are n observations, show that these maneuvers lead to a minorizing function

$$g(\theta | \theta^{(k)}) = -\frac{1}{p} \sum_{i=1}^{n} \frac{\exp(X_i^T \theta^{(k)})}{1 + \exp(X_i^T \theta^{(k)})} \sum_{j=1}^{p} \exp\{p X_{ij} (\theta_j - \theta_j^{(k)})\} + \sum_{i=1}^{n} Y_i X_i^T \theta = 0$$

Answer:

First,

$$\exp\{\mathbf{X}_i^T\boldsymbol{\theta}\} = \exp\Big\{\mathbf{X}_i^T\boldsymbol{\theta}^{(k)}\Big\}\exp\Big\{\mathbf{X}_i^T(\boldsymbol{\theta}-\boldsymbol{\theta}^{(k)})\Big\} = \exp\Big\{\mathbf{X}_i^T\boldsymbol{\theta}^{(k)}\Big\}\prod_{j=1}^{p}\exp\Big\{X_{ij}^T(\theta_j-\theta_j^{(k)})\Big\}$$

Next, using the arithmetic-geometric mean inequality,

$$\prod_{j=1}^{p}\exp\Big\{X_{ij}^T(\theta_j-\theta_j^{(k)})\Big\} = \prod_{j=1}^{p}\exp\Big\{pX_{ij}^T(\theta_j-\theta_j^{(k)})\Big\}^{\frac{1}{p}} \leq \frac{1}{p}\sum_{j=1}^{p}\exp\Big\{pX_{ij}^T(\theta_j-\theta_j^{(k)})\Big\}$$

Then

$$\exp\{\mathbf{X}_i^T\boldsymbol{\theta}\} \leq \exp\Big\{\mathbf{X}_i^T\boldsymbol{\theta}^{(k)}\Big\}\frac{1}{p}\sum_{j=1}^{p}\exp\Big\{pX_{ij}^T(\theta_j-\theta_j^{(k)})\Big\}$$

And $g(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$ is obtained by substituting the previous inequalities and the log likelihood from Problem 1.

(C) Finally, prove that maximizing $g(\theta|\theta^{(k)})$ consists of solving the equation

$$-\sum_{i=1}^{n}\frac{\exp(X_i^T\theta^{(k)})X_{ij}\exp(-pX_{ij}\theta_j^{(k)})}{1+\exp(X_i^T\theta^{(k)})}\exp(pX_{ij}\theta_j) + \sum_{i=1}^{n}Y_iX_{ij} = 0$$

for each j.

Answer:

Maximize $g(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$ by setting its derivative equal to 0 and solving.

$$\begin{aligned}
\nabla g(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}) &= \frac{\partial g(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})}{\theta_j} \\
&= \frac{\partial}{\theta_j}\left(-\frac{1}{p}\sum_{i=1}^{n}\frac{\exp(X_i^T\theta^{(k)})}{1+\exp(X_i^T\theta^{(k)})}\sum_{j=1}^{p}\exp\{pX_{ij}(\theta_j-\theta_j^{(k)})\} + \sum_{i=1}^{n}Y_iX_i^T\theta\right) \\
&= \frac{\partial}{\theta_j}\left(-\frac{1}{p}\sum_{i=1}^{n}\frac{\exp(X_i^T\theta^{(k)})}{1+\exp(X_i^T\theta^{(k)})}\sum_{j=1}^{p}\exp\{pX_{ij}\theta_j\}\exp\{-pX_{ij}\theta_j^{(k)}\} + \sum_{i=1}^{n}Y_iX_i^T\theta\right) \\
&= -\frac{1}{p}\sum_{i=1}^{n}\frac{\exp(X_i^T\theta^{(k)})}{1+\exp(X_i^T\theta^{(k)})}pX_{ij}\exp\{pX_{ij}\theta_j\}\exp\{-pX_{ij}\theta_j^{(k)}\} + \sum_{i=1}^{n}Y_iX_{ij} \\
&= -\sum_{i=1}^{n}\frac{\exp(X_i^T\theta^{(k)})X_{ij}\exp\{-pX_{ij}\theta_j^{(k)}\}}{1+\exp(X_i^T\theta^{(k)})}\exp\{pX_{ij}\theta_j\} + \sum_{i=1}^{n}Y_iX_{ij} \\
&= -\sum_{i=1}^{n}\pi^{(k)}X_{ij}\exp\{-pX_{ij}\theta_j^{(k)}\}\exp\{pX_{ij}\theta_j\} + \sum_{i=1}^{n}Y_iX_{ij}
\end{aligned}$$

Problem 3: simulation

Next we will implement logistic regression in R four different ways and compare the results using a short simulation study.

- implement using Newton's method from 1.1 in R
- implement using MM from 1.2 in R

- implement using `glm()` in R
- implement using `optim()` in R:
  - Use the option `method = "BFGS"`, which implements a Quasi-Newton approach

Simulation study specification:

- simulate from the model $logit(P(Y_i = 1|X_i)) = \beta_0 + \beta_1 X_i$
  - $\beta_0 = 1$
  - $\beta_1 = 0.3$
  - $X_i \sim N(0, 1)$
  - $n = 200$
  - $nsim = 1$
- For your implementation of MM and Newton's method, select your own starting value and stopping criterion, but make sure they are the same for the two algorithms

You only need to run the simulation using **one simulated dataset**. For each of the four methods, report:

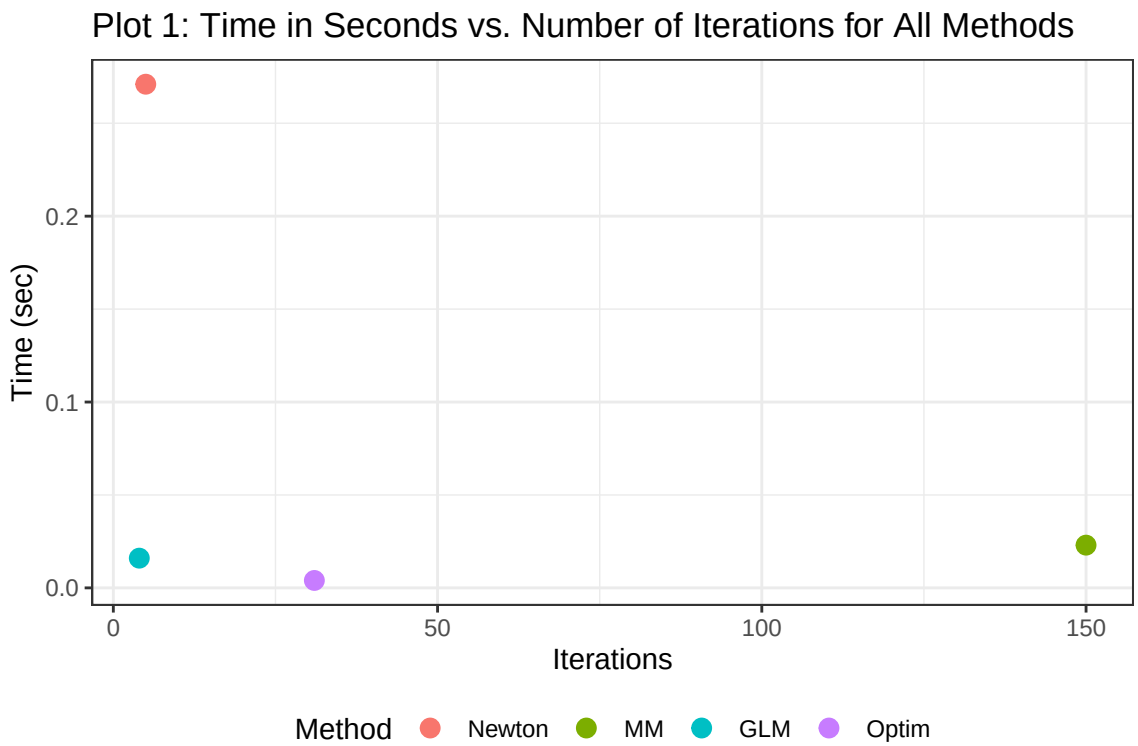
- $\hat{\beta}_0$, $\hat{\beta}_1$
- 95% confidence intervals for $\hat{\beta}_0$, $\hat{\beta}_1$
- computation time
- number of iterations to convergence

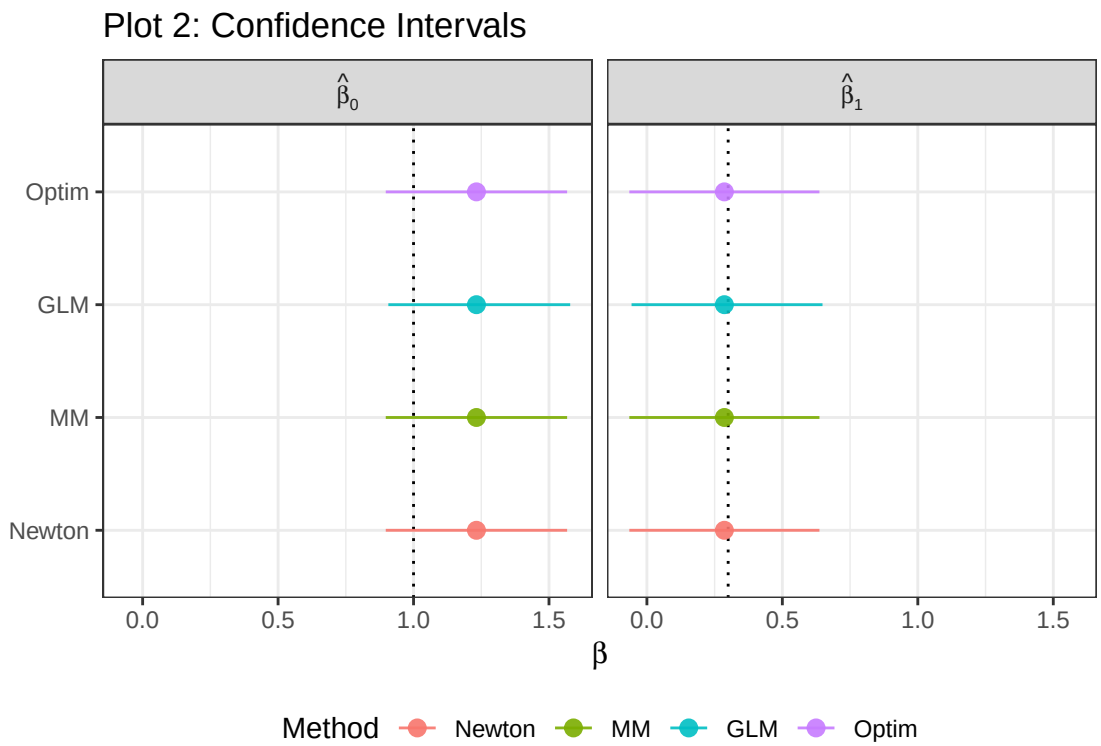
Make 2-3 plots or tables comparing your results, and summarize these findings in one paragraph.

Answer:

```r
# set seed
set.seed(03042026)

# load source files
source(here('source', 'logistic.R'))
source(here('source', 'logistic_simulate_data.R'))
source(here('source', 'logistic_methods.R'))

# simulate data
dat <- sim_data(beta0 = 1, beta1 = 0.3, n = 200)
attach(dat)

# run all methods
newton_res <- newton_method(X, Y)
mm_res <- mm_method(X, Y)
glm_res <- glm_method(X, Y)
optim_res <- optim_method(X, Y)
```

Method	$\hat{\beta}_0$	$\hat{\beta}_1$	Iterations	Time (sec)
Newton	1.232183	0.2859855	5	0.271
MM	1.232183	0.2859854	150	0.023
GLM	1.232183	0.2859855	4	0.016
Optim	1.232182	0.2859848	31	0.004





All methods converged and returned the same values of $\hat{\beta}_0$ and $\hat{\beta}_1$ with only minor rounding differences. Confidence intervals were also nearly identical. All methods slightly overestimated β_0 and underestimated but were still more accurate for β_1. The only difference seemed to be the computation time and number of iterations. Newton's method, which inverts the Hessian in every iteration, had few iterations but still took the longest at about a 3rd of a second at 0.352 seconds. Optim was the fastest of methods at 0.006 seconds, followed by GLM and then MM which had a comparable computation time (0.024 and 0.038 seconds, respectively). GLM had the fewest iterations at 4, followed by Newton at 5. MM had the required the

greatest number of iterations at 150, followed by Optim which took 31. From the plot, the computation time seemed unrelated to the number of iterations.

Problem 4: EM algorithm for censored exponential data

This will be a continuation of the lab problem on EM for censored exponential data. Suppose we have survival times $t_1, \ldots, t_n \sim Exponential(\lambda)$.

- Do not observe all survival times because some are censored at times $c_1, \ldots, c_n$.

- Actually observe $y_1, \ldots, y_n$, where $y_i = min(y_0, c_i)$
  - Also have an indicator δ_i where $\delta_i = 1$ is $y_i \leq c_i$
    - i.e. $\delta_i = 1$ if not censored and $\delta_i = 0$ if censored

Do the following:

- Derive an EM algorithm to estimate the parameter λ. Show your derivation here and report updates for the **E-step** and **M-Step**.
- Implement your EM in R and fit it to the `veteran` dataset from the `survival` package.
  - Report your fitted λ value. How did you monitor convergence?
  - Report a 95% confidence interval for λ, and explain how it was obtained.
  - Compare 95% confidence interval and results from those obtained by fitting an accelerated failure time model (AFT) in R with exponential errors. You can fit an AFT model using the `survreg()` function from the `survival` package. If you choose `dist = "weibull` and `shape = 1` as parameter arguments, this will provide exponential errors.

Answer:

Let $m = \sum_{i=1}^n \delta_i$. Then the complete data likelihood and log-likelihood are:

$$
\begin{aligned}
\hat{\lambda} &= \frac{n}{\sum_{i=1}^n t_i} \\
Q(\lambda|\lambda^{(k)}) &= \mathrm{E}\Big[l_C(\lambda|t_1, \ldots, t_n)|\mathbf{Y}, \lambda^{(k)}\Big] \\
&= n\log(\lambda) - \lambda\left[\sum_{i=1}^n \delta_i Y_i + (1-\delta_i)(Y_i + \frac{1}{\lambda^{(k)}})\right] \\
&= n\log(\lambda) - \lambda\left[\sum_{i=1}^n Y_i + \frac{1-\delta_i}{\lambda^{(k)}}\right] \\
&= n\log(\lambda) - \lambda\left\{\frac{n-m}{\lambda^{(k)}} + \sum_{i=1}^n Y_i\right\} \\
\frac{\partial Q(\lambda|\lambda^{(k)})}{\partial\lambda} &= \frac{n}{\lambda} - \frac{n-m}{\lambda^{(k)}} - \sum_{i=1}^n Y_i = 0 \\
\lambda^{(k+1)} &= \frac{n}{\frac{n-m}{\lambda^{(k)}} + \sum_{i=1}^n Y_i}
\end{aligned}
$$

E-step: $\frac{n-m}{\lambda^{(k)}} + \sum_{i=1}^n Y_i$

M-step: $\lambda^{(k+1)} = \frac{n}{\frac{n-m}{\lambda^{(k)}} + \sum_{i=1}^n Y_i}$

```
library(survival)
```

```r
Y <- as.integer(veteran$time)
delta <- as.integer(veteran$status)
```

```r
source(here('source', 'em_algorithm.R'))

# E-M algorithm
em_res <- em_alg(Y, delta, lambda0 = 0.001)

# survival package AFT
sfit <- survreg(Surv(time, status) ~ 1, dist = 'weibull', data = veteran)
lambda_aft <- exp(-coef(sfit))
ci_aft <- rev(exp(-confint(sfit)))
```

	$\hat{\lambda}$	2.5 %	97.5 %	CI Width
E-M Algorithm	0.007682	0.007671	0.007692	2.044498e-05
AFT Model	0.008286	0.006708	0.010236	3.527696e-03

The CI for the E-M algorithm was obtained using $\text{Var}\left(\hat{\lambda}\right) \approx \frac{\hat{\lambda}^2}{m}$. The CI was calculated as $\exp\left\{\log(\hat{\lambda}) \mp z_{\alpha/2} * \text{se}(\hat{\lambda})\right\}$.

Both methods yield similar estimates for λ with small confidence intervals. However, the E-M algorithm confidence interval was narrower.